Migrating SQL Server to Azure

Exhaustive inventory of targets, methods & tools | v2.9

17 August 2026

Contents

1 Why migrate in 2026 (the short "why now") 2

2 Taxonomy — separate targets, control planes, and methods 3

3 The Azure targets (8 families) 4

4 Control planes & assessment experiences 5
4.1 Microsoft tooling by source → target (assess · data · schema) 5

5 Migration methods per target 6
5.1 To SQL Server on Azure VM (IaaS rehost) 6
5.2 To Azure SQL Managed Instance (managed lift-and-shift) 6
5.3 To Azure SQL Database (cloud-native refactor) 7
5.4 To SQL database in Fabric *(GA target; Migration Assistant in Preview)* . . 8
5.5 To Containers / Arc-enabled SQL MI 8
5.6 Offline & network transfer accelerators (large estates) 8

6 Ancillary components (the cutover blockers) 9

7 Downtime strategy (cutover window) — the #1 architect criterion 9

8 Summary matrix — method / tool × target 10

9 Source-version & retirement reference 11

10 Cross-cloud sources & reverse migration 13

11 Third-party alternatives (when they beat the native stack) 14

12 Decision criteria & "when to recommend what" 14

13 Field insights — recurring pitfalls 15

14 FY27 SQL Motion context & AI Migration Agent 17

15 Commercial levers & funding programs (FY27) 17
15.1 Commercial / licensing levers (permanent TCO reducers) 18
15.2 Microsoft-funded engagement programs 18
15.3 Partner mechanics 18

16 Sources (Microsoft Learn) 19

17 Document version & changelog 21

18 Appendix — Using the `sql-migration-advisor` Copilot CLI skill 24
18.1 Install (once) 24
18.2 Example output 24

Goal. Exhaustively list every way and every tool to migrate a SQL Server database to an Azure service (all PaaS, including containers) or a VM (Azure VM / Azure VMware Solution).

Audience. Partners, architects, and customer DBAs — usable in pre-sales and as the knowledge base behind the *SQL in a Day* AI Migration Agent (§14).

Verification. Tool retirements, version requirements and target families were cross-checked against Microsoft Learn and product announcements (current as of August 2026). Links are gathered in §16 Sources.

Version. v2.9 — 17 August 2026. Change history in §17 Document version & changelog.

IMPORTANT 2025–2026 tooling reset — read this first. - **Data Migration Assistant (DMA)** is RETIRED (16 July 2025). - **Azure Data Studio (ADS)** is RETIRED (28 February 2026). The Azure SQL Migration extension has no separately announced retirement date; do not anchor new runbooks on it because the replacement experiences are SSMS 22, Azure Arc migration and modern DMS. - The new entry point is the **SSMS 22 Migration Component** (assess + migrate from SSMS), complemented by SQL Server migration in Azure Arc (portal, with Copilot — now **GA to both Azure SQL MI and SQL Server on Azure VM**) and Azure Database Migration Service (DMS). - Modern **DMS** supports offline-only migration to Azure SQL Database (online / minimal-downtime is available for Managed Instance and SQL VM targets, per the DMS supported-scenarios matrix). - **Azure DMS *classic*** SQL Server scenarios have been absorbed into the current DMS portal experience — use modern DMS (an Azure resource, via portal / PowerShell / CLI).

---

1 Why migrate in 2026 (the short "why now")

- **AI runs on data.** Modern, managed databases are the foundation; SQL Server 2025 / Azure SQL add a native `vector` type + functions, **DiskANN** vector indexing (high-throughput ANN search), native `json`, and zero-ETL **Fabric Mirroring** (analytics on OneLake with no pipelines) — making the estate AI-ready and improving migration (enhanced distributed AG).
- **End-of-support pressure.** Out-of-support SQL Server/Windows versions push modernization; the SQL Server Extended Security Updates programme now covers SQL Server 2014 and SQL Server 2016 only, and ESU is free on Azure VMs / AVS for SQL Server 2014. SQL Server 2016 is paid everywhere (including Azure VM) and non-Azure environments use Azure Arc for ESU subscription/billing — this changes the *stay vs migrate* math.
- **Cost levers.** Azure Hybrid Benefit (AHB) applies to Azure SQL Database General Purpose / Business Critical in the vCore provisioned compute tier, Managed Instance and VM (not Fabric SQL DB, DTU or serverless). Hyperscale is a creation-date cohort: only single databases with provisioned compute created before 15 December 2023 can apply AHB, and only until December 2026; databases created on or after that date have no SQL software license fee to offset.

- **Partner leverage.** Sharing a deal with a partner increases win rate and deal size; programs like Cloud Accelerate Factory and SQL in a Day industrialize delivery (§14). Commercial & funding levers are detailed in §15.
- **Concrete forcing function.** SQL Server 2016 left extended support on 15 July 2026 and is now in ESU Year 1 (15 Jul 2026–13 Jul 2027; Year 3 ends 17 Jul 2029). "Stay" means paid ESU, and SQL Server 2016 requires a paid ESU subscription even on Azure VM.

Version	Extended End of Support	Status (Jul 2026)
SQL Server 2012	12 Jul 2022 — ESU ended Jul 2025	✗ Out of support
SQL Server 2014	10 Jul 2024 — ESU Y3 ends 13 Jul 2027	Out of support (ESU window)
SQL Server 2016	15 Jul 2026 — ESU Y1 15 Jul 2026–13 Jul 2027; Y3 ends 17 Jul 2029	Out of support — ESU Y1
SQL Server 2017	13 Oct 2027	✓ Extended support
SQL Server 2019	9 Jan 2030	✓
SQL Server 2022	12 Jan 2033	✓
SQL Server 2025	GA 18 Nov 2025 — extended end 7 Jan 2036	✓ Latest

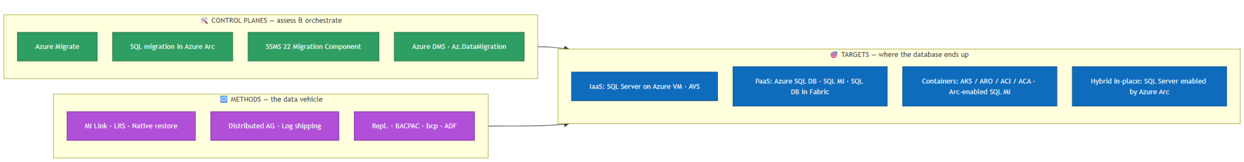
Dates are Microsoft Lifecycle Extended End Dates, quoted as published (2014, 2016, 2017, 2019, 2022, 2025). Microsoft records them at 06:59:59 PT on the stated day, so the last covered update ships on the Patch Tuesday before it — 14 July 2026 for SQL Server 2016. Quote the Lifecycle date in a customer conversation, and the Patch Tuesday only if someone asks which update was the last.

Windows Server often sits under SQL Server — **WS 2012/2012 R2** is out of support (ESU to Oct 2026), WS 2016 EOS ~Jan 2027. Factor the OS into the move.

---

2 Taxonomy — separate targets, control planes, and methods

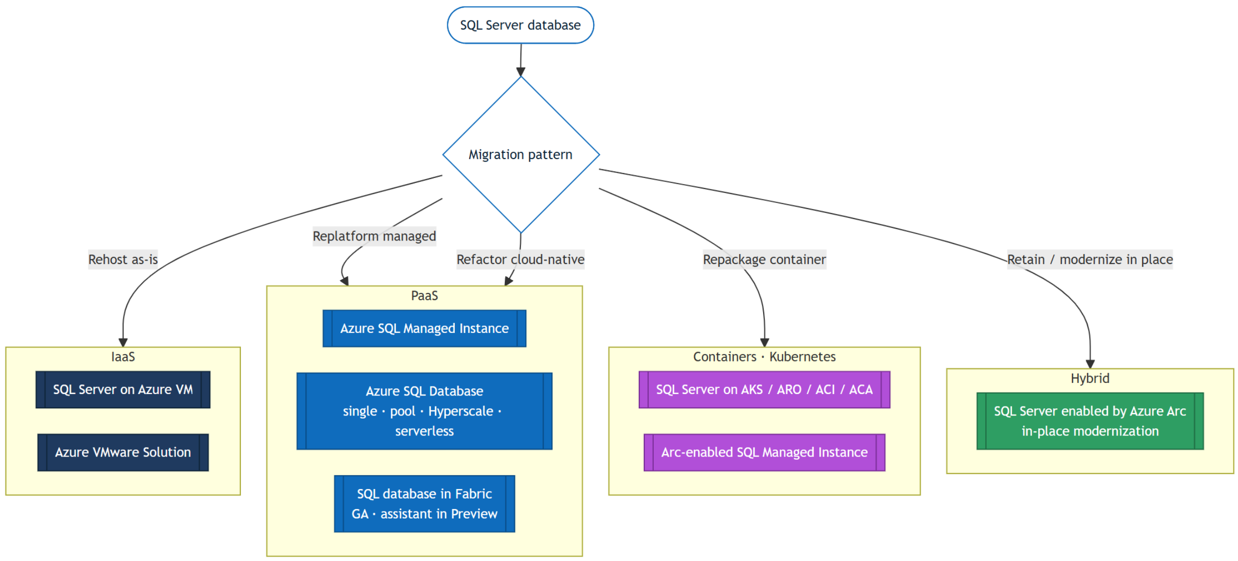
The single most common mistake is to mix these three layers. Keep them distinct:



- **Targets** = runtime destinations (where the DB lives).
- **Control planes / experiences** = how you assess, recommend and orchestrate (Azure Migrate, Arc, SSMS 22, DMS). Azure Arc-enabled SQL *Server* is a control plane, not a runtime target.
- **Methods** = the actual data-movement vehicle (MI Link, LRS, backup/restore, replication…).

---

3 The Azure targets (8 families)



#	Target	Layer	When to choose it	Compatibility	Doc
1	SQL Server on Azure VM	IaaS	Faithful lift & shift: OS / file-system control, exact version, FileStream/FileTable, PolyBase, cross-instance DTC, third-party agents.	Full	overview
2	Azure VMware Solution (AVS)	IaaS	Zero-refactor data-center exit for existing VMware estates; keeps FCI and Always On AG; migrate with VMware HCX / vMotion.	Full	AVS
3	Azure SQL Managed Instance	PaaS	Managed lift-and-shift: keep instance objects (logins, SQL Agent, server triggers, cross-DB, linked servers), native vNet. Tiers GP / BC / Next-gen GP *(GA since Nov 2025 — Elastic SAN backend: 500 DBs, 128 vCores, 32 TB, 80K IOPS, 192 MB/s log, 3–4 ms latency, configurable IOPS/memory; ~5x better price-per-DB by density)*. Free offer: 1 instance / 12 months. No Hyperscale on MI.	~Near-full (instance)	overview
4	Azure SQL Database	PaaS	Cloud-native app / microservice. Models: single DB / elastic pool; tiers GP / BC / Hyperscale; purchasing vCore / DTU / serverless. Hyperscale scales to 128 TB (large / HTAP); serverless for intermittent; elastic pools for consolidation. Free offer: 10 serverless DBs for the subscription lifetime.	Database surface (no instance-level)	overview
5	SQL database in Fabric	PaaS	Fabric-native OLTP unified with OneLake. The **target itself is GA**; only the Fabric Migration Assistant is Preview, with tool limits (DACPAC schema ≤ 20 MB, on-prem data gateway only, no Private Link). The target also accepts T-SQL, transactional replication, Fabric pipelines / Data Factory copy jobs, Dataflow Gen2, and other TDS-capable tools, so assistant limits alone must not eliminate it. Assess the database surface before committing an enterprise OLTP workload.	Subset of the SQL Server surface	SQL database in Fabric · Migration Assistant *(Preview)*
6	SQL Server in containers — AKS / ARO / ACI / ACA	Container	Full control of the engine in a container (dev/test, edge, custom). Pod + PersistentVolume; HA via the Kubernetes scheduler.	High — SQL on Linux (no FILESTREAM/FileTable, SSRS/SSAS/SSIS, ML Services; SQL Agent off by default)	SQL on Kubernetes
7	Azure Arc-enabled SQL Managed Instance	Container (PaaS)	Managed SQL MI engine on any Kubernetes (AKS, ARO, EKS, GKE, OpenShift) via `kubectl` +CRD. Sovereignty / edge / multi-cloud.	~Same as SQL MI	create Arc SQL MI
8	SQL Server enabled by Azure Arc	Hybrid (control plane)	Not a runtime target — the bootstrap pillar: connect on-prem SQL *without touching it* → weekly continuous assessment, blocker detection, ESU, PAYG (OpEx) licensing, governance, and a portal Copilot-assisted Database migration — to **Azure SQL MI** (MI Link / LRS) and, **GA since July 2026**, to **SQL Server on Azure VM** (native backup/restore). Arc-enabled SQL Server 2014+ overall; path-specific floors apply (MI Link: SQL Server 2016+, Windows Server only on this Arc path; use the conservative 2014+ Arc floor for LRS because Microsoft Learn is inconsistent — see §9).	n/a	Arc migration

Lift & shift is not VM-only. Both SQL Server on Azure VM (IaaS) and Azure SQL Managed Instance (managed PaaS) are valid lift-and-shift targets. Since 2024, MI compatibility (SQL Agent, cross-DB queries, linked servers…) makes it the default *managed* lift-and-shift.

4 Control planes & assessment experiences

Tool / experience	Role	Status (2026)	Notes
Azure Migrate	Discovery / assessment / sizing / business case at scale	GA (+ Arc-based agentless discovery, Preview)	Appliance (VMware/Hyper-V/Physical) or import-based or Arc-based. Right-sizes SQL DB / MI / VM.
SQL Server migration in Azure Arc	Portal-driven assess + migrate for any Arc-enabled SQL Server	GA; **MI and VM targets both GA**	Copilot-assisted; targets **Azure SQL MI** (MI Link / LRS) and — **GA July 2026** — **SQL Server on Azure VM** (lift-and-shift, native backup/restore); continuous assessment; Arc-enabled SQL Server 2014+ overall, with MI Link requiring SQL Server 2016+ and, on this Arc-driven path, **Windows Server only**. The Azure Arc portal MI wizard can select up to 10 databases per batch with Azure Extension for SQL Server ≥ 1.1.3348.364 (earlier versions: one database at a time); this is an Arc wizard batch limit, not MI Link capacity.
SSMS 22 Migration Component	DBA-first entry point: assess + launch a recommended migration path from SSMS	GA (Windows-only)	Replaces DMA-era workflows and complements the ADS migration extension. **Migrate SQL Server assesses SQL Server instances and migrates them to Azure SQL today.** Backup/restore, MI Link, DMS. Arc-enabled sources can reuse readiness assessments already collected through Azure Arc.
Azure DMS (modern)	Managed migration orchestration (Azure resource · portal / PowerShell / CLI)	GA	Use the modern DMS — former DMS *classic* SQL scenarios are absorbed into the current portal experience. Offline-only to Azure SQL DB; online/minimal-downtime to MI / SQL VM (MI Link preferred for MI).
PowerShell `Az.DataMigration` / Azure CLI	Automate DMS at scale (CI/CD)	GA	Often the only viable path beyond ~50 databases.
SSMA	Heterogeneous conversion (schema/code/data)	GA	Oracle / Sybase / DB2 / MySQL / Access → Azure SQL. Not for homogeneous SQL→SQL.
Database Experimentation Assistant (DEA)	Capture + replay a production workload on the target to validate performance *before* cutover	Retired 15 December 2024	Replaced by a documented validation stack: capture with Extended Events (SQL Trace/Profiler deprecated), replay with RML Utilities / OStress, then analyse with Query Store and DMVs. The *concept* (catch plan/compat regressions before cutover) remains valuable; the MS tool itself is retired.

NOTE Retired — do not use in new runbooks: DMA (16 Jul 2025) and Azure Data Studio itself (28 Feb 2026). The Azure SQL Migration extension has no separately announced retirement date, but migration work should continue via SSMS 22 / Azure Arc / modern DMS / `Az.DataMigration` CLI. SQL Data Sync retires 30 Sep 2027 — don't build new sync/migration on it (use ADF, transactional replication or AG).

4.1 Microsoft tooling by source → target (assess · data · schema)

Common tooling per source/target combination (homogeneous and heterogeneous); third-party CDC options are labelled separately from current Microsoft migration guidance:

Source	Target	Assess	Data migration	Schema
SQL Server (Arc-enabled)	Azure SQL MI	SQL migration in Azure Arc	SQL migration in Azure Arc	Not needed
SQL Server (Arc-enabled)	SQL Server on Azure VM	SQL migration in Azure Arc	SQL migration in Azure Arc (native backup/restore)	Not needed
SQL Server (not Arc)	Azure SQL VM / MI	Azure Migrate	DMS	Not needed
SQL Server	Azure SQL DB	Azure Migrate	STRIIM (online) · DMS (offline)	DMS
Sybase	Azure SQL	Azure Migrate	STRIIM	SSMA for Sybase
Oracle	Azure SQL	Azure Migrate	STRIIM	SSMA for Oracle

GitHub Copilot is applicable as AI-assisted migration tooling (schema/code conversion inside SSMA; app-level via GitHub Copilot App Modernization for

.NET / Java). Portfolio & application/code assessment partners complementary to Azure Migrate: Dr Migrate, CAST Highlight, UnifyCloud.

---

5 Migration methods per target

Standardized columns (Microsoft Learn style): **Method · Min source · Target/min · Downtime · Key constraints**.

5.1 To SQL Server on Azure VM (IaaS rehost)

Method	Min source	Downtime	Key constraints / notes
Azure Migrate (lift & shift)	SQL 2008 SP4	Online (replication)	Whole VM/instance, incl. FCI and AG; up to ~35,000 VMs.
SQL migration in Azure Arc → SQL VM	SQL 2014+ (Arc-enabled)	Offline (backup/restore)	**GA (July 2026):** portal-driven, Copilot-assisted lift-and-shift for Arc-enabled sources — provisions the target Azure VM and runs native backup/restore. A phased on-ramp: rehost now, modernize to SQL MI / SQL DB later.
Distributed availability group (DAG)	SQL 2016	Near-zero	Reuse on-prem AG; needs AD Domain Services (or workgroup AG + certs) and ports open.
Backup to a file (.bak) + copy	SQL 2008 SP4	Offline	Simple, supports > 1 TB; use compression / multi-file split for WAN.
Backup to URL (Azure Blob)	SQL 2012 SP1 CU2	Offline	SQL 2012 SP1 CU2 / 2014: page blob + storage-account credential, 1 TB max. SQL 2016+: block blob + SAS credential, 12.8 TB via striping. For > 1 TB on 2012/2014 use local backup + AzCopy.
Detach & attach (MDF/LDF via Blob)	SQL 2008	Offline	For very large DBs where backup/restore is too slow.
Log shipping	SQL 2008	Minimal	Windows-only (not for SQL on Linux sources).
Always On AG	SQL 2012	Near-zero	Fail an existing AG onto Azure VM replicas.
Convert machine to VHD / Ship hard drive / Azure Data Box	any	Offline	Estate exit with limited WAN; multi-TB `.bak` / `.bacpac` via Data Box.

5.2 To Azure SQL Managed Instance (managed lift-and-shift)

Method	Min source	Downtime	Key constraints / notes
Managed Instance link (MI Link)	SQL 2016 and later (incl. 2022, 2025; Ent/Std/Dev; host OS supported by that SQL version — Windows Server throughout, Linux from SQL 2017, SQL 2016 Windows-only)	Near-zero (online; <1 minute cutover)	Distributed-AG based; R/O readable target during migration; reverse failback to SQL 2022 / 2025 (DR / Azure exit); one link per database, up to 100 links on General Purpose / Business Critical and up to 500 links on Next-gen General Purpose; requires MI Link network ports below.
Log Replay Service (LRS)	Standalone LRS: SQL Server 2008–2022	Online sync; cutover downtime	Full/diff/log → Azure Blob; public endpoint; 30-day max window; target remains RESTORING / NORECOVERY (no reads/writes) until cutover. Cutover restores the final backup: minutes on GP with a small final backup, but potentially hours on Business Critical while replicas are seeded. Supports up to the service-tier database limit (for example 100 GP, 500 Next-gen GP), with 100 simultaneous restores per instance and 150 per subscription. Arc portal migration uses the conservative 2014+ Arc floor; standalone LRS remains 2008–2022.
Native backup & restore (.bak)	SQL 2008	Offline	Simplest; migrate TDE certificate *before* restore or it fails late; master/msdb restore not supported (script instance objects).
Transactional replication	SQL Server 2016+	Online	MI can be a subscriber from SQL Server 2016 and later; exact publisher/distributor/subscriber combinations depend on the MI update policy, so check the supportability matrix.
bcp	any	Offline	High-speed data-only / partial.
Smart Bulk Copy	any	Offline	Archived Azure-Samples parallel-copy wrapper over bulk copy; data-only / partial.
BACPAC / SqlPackage	any	Offline	Smaller DBs / simple.
Azure Data Factory — Copy	any	Offline / batch	When migration = integration / transformation.

MI Link network ports. Always open the full documented set, regardless of service tier, update policy, VPN / ExpressRoute / VNet peering, or other connectivity mechanism: on the **SQL Managed Instance subnet NSG**, inbound **5022** and **11000–11999** from the SQL Server IP, and outbound **5022** to the SQL Server IP (Microsoft's summary table states 5022 + 11000–11999 both inbound and outbound for the MI NSG). On the **SQL Server host OS firewall and any corporate firewall**, allow inbound **5022** from the MI subnet /24 and outbound **5022** and **11000–11999** to the MI subnet. Ports **11000–11999** carry the MI-side distributed-AG HADR data-replication channel; the actual MI HADR port is dynamically assigned within the range and can be checked with `sys.dm_hadr_fabric_config_parameters` where `parameter_name = 'HadrPort'` (for example, 11002). MI-side ports cannot be customized; the SQL Server-side endpoint port can. Microsoft's simplified SQL Server-side table lists only 5022, but the authoritative prose also requires outbound 11000–11999 — follow the prose.

MI compatibility nuance — PolyBase and DTC. PolyBase used only for cloud file access (Azure Blob Storage / ADLS Gen2, CSV/delimited or Parquet through `OPENROWSET(BULK)`, `CREATE EXTERNAL TABLE`, CETAS) is **MI eligible**. PolyBase used to query Oracle, Teradata, MongoDB or another SQL Server through external RDBMS connectors is **MI ineligible**; use SQL Server on Azure VM or Synapse. Homogeneous SQL↔SQL DTC (SQL MI ↔ SQL MI / SQL Server) is **MI eligible** via managed DTC; heterogeneous DTC to a third-party RDBMS is **MI ineligible** and needs SQL VM or refactoring. FILESTREAM / FileTable remains a hard blocker for SQL MI and SQL DB.

5.3 To Azure SQL Database (cloud-native refactor)

Method	Min source	Downtime	Key constraints / notes
Azure DMS (offline)	SQL 2008+	Offline	Offline only to Azure SQL DB (online/minimal-downtime is available for MI / SQL VM, not SQL DB).
STRIIM (online CDC)	any	Online (near-zero)	Microsoft-recommended online / CDC data migration to Azure SQL DB — fills the gap that DMS is offline-only for SQL DB; pair with DMS for schema.
Transactional replication	SQL Server 2016 and later (incl. 2022, 2025)	Online	Azure SQL Database can be a push subscriber only, with snapshot and one-way transactional replication. Peer-to-peer and merge are not supported; replicated tables require primary keys. Article limits include unsupported conversions for `hierarchyid`, FILESTREAM and spatial to MAX types, plus partitioning/index feature limits.
BACPAC / SqlPackage	any	Offline	Small/medium; `SqlPackage` for scale.
bcp	any	Offline	Data-only / bulk.
Smart Bulk Copy	any	Offline	Archived Azure-Samples parallel-copy wrapper; data-only / bulk.
Azure Data Factory — Copy	any	Offline / batch	BI / integration.

✗ **Not supported to Azure SQL Database:** native .bak restore, detach/attach, MI Link. SQL Agent → use Elastic Jobs.

DACPAC vs BACPAC. A DACPAC packages the schema only; a BACPAC packages schema + data. Both are produced/consumed by `SqlPackage` and the VS Code MSSQL extension — a simple way to migrate (BACPAC) or version (DACPAC) offline.

5.4 To SQL database in Fabric *(GA target; Migration Assistant in Preview)*

Method	Downtime	Key constraints / notes
Fabric Migration Assistant — DACPAC	Offline	Preview tool limits: schema via DACPAC ≤ 20 MB; AI-assisted compatibility fixes; data via Fabric Data Factory copy job + on-prem data gateway only (no VNet gateway / Private Link for the assistant). These are Migration Assistant limits, not target-wide Fabric SQL database limits.
Transactional replication	Online	Fabric SQL database can be a push subscriber only. Publishing to Fabric SQL database requires SQL Server 2022 RTM CU12 or greater; snapshot and one-way transactional replication are supported, peer-to-peer and merge are not, and replicated tables require primary keys.
bcp	Offline	Documented against Fabric SQL database "just like any other SQL Database Engine product", and *SQL database in Microsoft Fabric* appears in the bcp utility **Applies to** banner. Fabric SQL database accepts no SQL authentication: connect with Microsoft Entra ID using `-G`, against `<server>.database.fabric.microsoft.com,1433`. Data-only — pair with a schema method.
Fabric Data Factory — Copy activity / Copy job / Dataflow Gen2	Offline / batch	Source **and** destination, Beta. Organizational-account authentication; None / on-premises / virtual-network gateway. **Azure** Data Factory has no Fabric SQL database connector — it ships Fabric Lakehouse and Fabric Warehouse only, so an Azure Data Factory pipeline cannot target this database.
T-SQL / other TDS-capable tools	Offline / batch	Alternative ingestion paths into Fabric SQL database; do not eliminate the target solely because Migration Assistant Preview limits do not fit.
Fabric Mirroring for SQL Server	n/a (continuous)	NOT a one-shot migration — near-real-time CDC replication to OneLake for analytics. GA (Nov 2025), optimized for SQL Server 2025. Complementary "analytical modernization" path.

5.5 To Containers / Arc-enabled SQL MI

Target	Method	Downtime	Notes
Arc-enabled SQL MI (AKS/ARO/…)	Native backup/restore, point-in-time restore	Offline	Requires Arc data controller. Use native backup/restore; use another method only where explicitly supported for Azure Arc-enabled SQL MI. **Managed Instance link is not supported for this target** — it is scoped to Azure SQL Managed Instance, and exposing a SQL endpoint does not change that (see the §8 footnote).
SQL Server container (mcr image)	Backup/Restore via mounted volume, detach/attach, BACPAC, bcp, ADF	Offline / Online (repl.)	Persist on Azure Disk (AKS/ARO) / Azure Files (ACI/ACA); HA via scheduler.

WARNING Containers are not a fully-managed cloud database. Plain `mssql` containers (AKS / ARO / ACI / ACA) put patching, high availability and backups entirely on you — best suited to dev/test and edge scenarios. Arc-enabled SQL MI automates engine patching, backups and HA through the Arc data controller, but you still own the Kubernetes cluster, persistent storage and DR. Neither is equivalent to the fully-managed Azure SQL Database / Managed Instance PaaS.

5.6 Offline & network transfer accelerators (large estates)

For multi-terabyte databases the network — not the method — is usually the bottleneck. Plan the **bulk transfer / initial seed** explicitly:

Accelerator	Use it for	Notes
Azure Data Box / Data Box Heavy	One-shot physical move of multi-TB `.bak` / `.bacpac` or a whole VM fleet	Data Box ≈ 80 TB usable; Data Box Heavy ≈ 770 TB for hundreds of TB. Microsoft recommends Data Box to migrate a SQL server/estate and as the initial *seed* before syncing the delta over the network.
ExpressRoute / dedicated circuit	Sustained high-throughput online transfer & seed	Avoids WAN time-outs on long-running backup/restore, LRS or replication.
AzCopy + Backup-to-URL	Push large local backups to Blob	Required above 1 TB **only on SQL 2012 SP1 CU2–2014**, where Backup to URL writes a page blob capped at 1 TB. On SQL 2016+ Backup to URL uses block blobs and reaches 12.8 TB with striping, so local backup + AzCopy is a throughput and retry choice there, not a size limit.
Azure Storage Mover	Orchestrated bulk file transfer on-prem → Azure Files / Blob	Centralized, resumable; good for backup repositories.
Compressed / split / encrypted backups	Shrink & secure the payload in transit	Compression + multi-file split optimize WAN; backup encryption (SQL 2014+) for compliance during transit.

Seed-then-sync pattern. For very large or low-downtime migrations: ship the initial full backup via Data Box (or AzCopy over ExpressRoute), then catch up the delta with log restore (LRS), MI Link, transactional replication or log shipping before cutover.

---

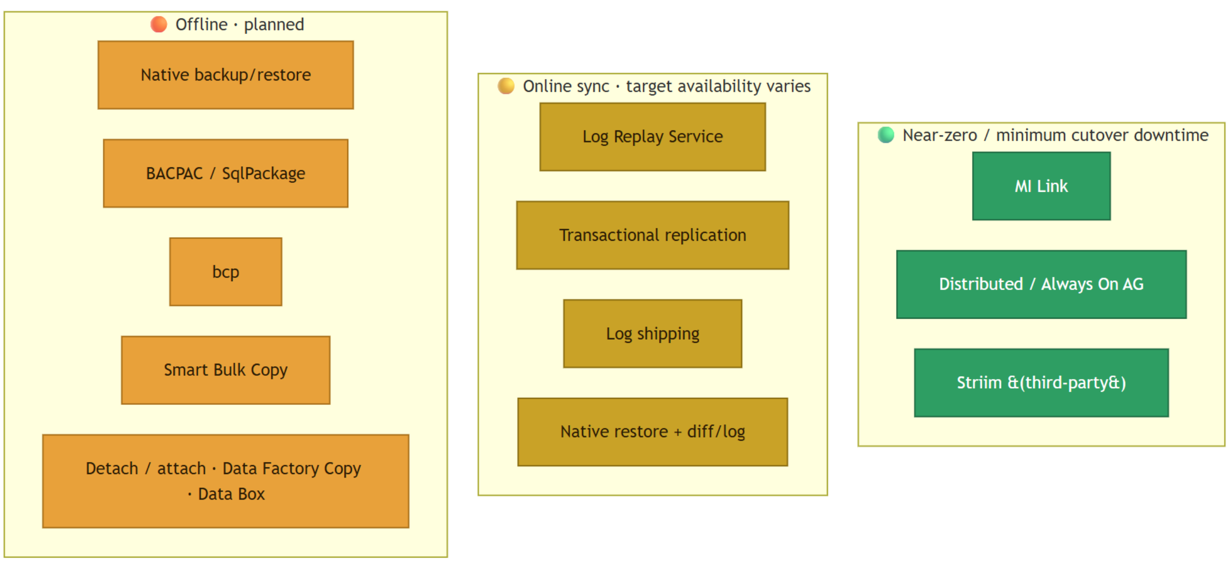
6 Ancillary components (the cutover blockers)

Databases rarely move alone — these sub-components block go-live if forgotten:

Component	Path to Azure
SQL Agent jobs	MI: native · SQL DB: Elastic Jobs
Logins / users	Script + recreate; DMS does not migrate Windows logins by default (enable option + grant MI read access to Entra ID).
SSIS	Azure-SSIS Integration Runtime (SSISDB via DMS).
SSRS	**Power BI paginated reports** for the managed cloud migration of RDL workloads; **Power BI Report Server on a VM** when the managed service does not fit. Starting with SQL Server 2025 (17.x), on-premises reporting services is consolidated under Power BI Report Server; no new SSRS versions after SSRS 2022, which follows the SQL Server 2022 lifecycle and is supported until 12 Jan 2033.
SSAS	Azure Analysis Services or Power BI Premium (XMLA).
Linked servers / cross-DB	Supported on VM/MI; not on SQL DB — refactor required.
TDE	Migrate the server-level certificate *before* any native restore to MI.

---

7 Downtime strategy (cutover window) — the #1 architect criterion



Separate **target availability during sync** from **business cutover downtime**; calling every staged method simply “offline” hides the operational difference:

Method	`targetAvailabilityDuringSync`	`businessCutoverDowntime`
MI Link	read-only (secondary queryable)	< 1 minute
LRS	unavailable (RESTORING/NORECOVERY)	minutes (GP, small final backup) → **hours** (Business Critical, replica seeding)
Native backup/restore	not present	full restore time
Transactional replication	read-write (subscriber accessible)	near-zero
DMS offline	not present	total migration execution time

Microsoft describes LRS as an online migration with expected downtime during cutover, not as a true minimum-downtime method. Prefer MI Link for Business Critical when sub-minute cutover is required because MI Link is the true online option and its secondary is queryable during sync.

Sizing rule. Never size MI/SQL DB on *average* CPU. Use a Perfmon baseline ≥ 7 days + ~20% headroom for cutover. Network: don't estimate `size ÷ bandwidth` — Backup-to-URL throughput is capped by the Blob layer; test with AzCopy + one full backup before committing a go-live date, and plan a rollback / fallback window in case the estimate proves optimistic.

---

8 Summary matrix — method / tool × target

Method / tool	SQL VM	AVS	SQL MI	SQL DB	Fabric SQL DB	Arc SQL MI	SQL container
Azure Migrate (assess)	✓	✓	✓	✓	–	–	–
DMS	✓	–	✓ (online)	✓ (offline)	✗	–	–
MI Link	reverse	✗	✓	✗	✗	–[3]	✗
Log Replay Service	✗	✗	✓	✗	✗	–	✗
Native backup/restore	✓	✓	✓	✗	✗	✓	✓
Distributed / Always On AG	✓	✓	✗	✗	✗	✗	✗
HCX / vMotion	✗	✓	✗	✗	✗	✗	✗
Transactional replication	✓	✓	✓	✓[1]	✓[2]	✓	✓
BACPAC / SqlPackage	✓	✓	✓	✓	✓ (DACPAC)	✓	✓
bcp	✓	✓	✓	✓	✓[4]	✓	✓
Smart Bulk Copy	✓	✓	✓	✓	–	–	–
Data Factory Copy	✓	✓	✓	✓	✓[5]	✓	✓
Fabric Migration Assistant	✗	✗	✗	✗	✓	✗	✗
SSMA (heterogeneous)	✓	✓	✓	✓	–	✓	✓

✓ supported · ✗ n/a · – indirect/non-first-class · reverse only · [1] SQL DB transactional replication: SQL Server 2016+ publisher, push subscriber only; snapshot and one-way transactional only. · [2] Fabric SQL database transactional replication requires SQL Server 2022 RTM CU12+ publisher, push subscriber only. · [3] Managed Instance link is scoped to **Azure SQL Managed Instance**, not Azure Arc-enabled SQL Managed Instance. Exposing a SQL endpoint on the Arc target does not make MI Link available; use native backup/restore or MI-compatible data movement. · [4] **bcp** lists *SQL database in Microsoft Fabric* in its own **Applies to** banner, and Fabric documents a dedicated Connect with bcp utility procedure — "just like any other SQL Database Engine product". Fabric SQL database accepts no SQL authentication, so the connection must use Microsoft Entra ID with the `-G` option. **Smart Bulk Copy** is a separate archived community sample and claims no Fabric, Arc SQL MI or container support; it is split out of this row for that reason. · [5] Fabric SQL database is served by **Fabric** Data Factory, whose SQL database connector supports Copy activity, Copy job and Dataflow Gen2 as both source and destination (Beta; organizational-account authentication; None / on-premises / virtual-network gateway). **Azure** Data Factory publishes Fabric Lakehouse and Fabric Warehouse connectors but **no Fabric SQL database connector** — do not plan an Azure Data Factory pipeline against this target.

---

9 Source-version & retirement reference

Tooling timeline: DMA retired Jul 2025 → DEA retired 15 December 2024 → ADS retired Feb 2026 (Azure SQL Migration extension: no separate announced retirement) → former DMS *classic* SQL scenarios absorbed into the current DMS portal experience → SQL Data Sync retires Sep 2027.

Pre-cutover workload validation: capture with Extended Events (SQL Trace/Profiler deprecated), replay with RML Utilities / OStress (updated Oct 2024 to work over ODBC/OLE DB without SNAC), and analyse with Query Store + DMVs. SQL Server Distributed Replay is deprecated as of SQL Server 2022 and is not available in SQL Server 2022 or later; the Controller was removed from Setup and the feature depended on SNAC, which was removed.

Item	Status / requirement
DMA	Retired 16 Jul 2025 → SSMS 22 / Arc / Azure Migrate
Azure Data Studio / SQL Migration extension	ADS retired 28 Feb 2026; migration extension has no separate announced retirement → VS Code + MSSQL for development; SSMS 22 / DMS for migration
DMS *classic* — SQL Server scenarios	Absorbed into the current DMS portal experience → modern DMS (portal / PowerShell / CLI)
Database Experimentation Assistant (DEA)	Retired 15 December 2024 → capture with Extended Events, replay with RML Utilities / OStress, analyse with Query Store + DMVs
Distributed Replay	Deprecated as of SQL Server 2022 and not available in SQL Server 2022 or later → use RML Utilities / OStress
SQL Data Sync	Retires 30 Sep 2027 → use ADF / transactional replication / AG
DMS → Azure SQL DB	Offline only (online available for MI / SQL VM)
MI Link source	SQL Server 2016 and later (incl. 2022, 2025), Ent/Std/Dev editions, on a host OS supported by that SQL Server version: Windows Server throughout, and **Linux from SQL Server 2017** onwards (SQL Server 2016 is Windows Server only). Windows hosts must be **Windows Server 2012 or later**: Microsoft states this in the link Limitations, because Windows 10 and 11 clients cannot enable the Always On availability group feature the link depends on. Requires sysadmin, distributed AG capability, AG endpoints, MI Link network ports (5022 plus MI-side HADR 11000–11999) and connectivity to the MI VNet
Standalone LRS source	SQL Server 2008–2022; SQL Server 2012 SP1 CU2+ can `BACKUP TO URL` directly (page blob to 1 TB on 2012/2014, block blob + SAS to 12.8 TB on 2016+), older builds need local backup + upload
Azure Arc portal migration source floors	Arc-enabled SQL Server 2014+ overall; MI Link method requires SQL Server 2016+ and, on this Arc-driven path specifically, **Windows Server only** (MI Link configured outside the Arc portal also supports Linux from SQL Server 2017); Microsoft documentation inconsistency: the Arc MI method table says LRS method SQL Server 2012+ / Windows Server 2012+, but the same page and the overall Arc migration overview state SQL Server migration in Azure Arc starts with SQL Server 2014 (12.x), so use the conservative 2014+ Arc experience floor. Standalone LRS outside Arc remains SQL Server 2008–2022.
Native restore → MI	SQL Server 2008+
Transactional replication → MI	SQL Server 2016+; exact combination depends on MI update policy/supportability matrix
Transactional replication → SQL DB / Fabric SQL DB	SQL Server 2016+ publisher for Azure SQL DB; Fabric SQL database publishing requires SQL Server 2022 RTM CU12+
Backup-to-URL minimum source	SQL Server 2012 SP1 CU2 (page blob + storage-account credential); block blob + SAS from SQL Server 2016
Backup-to-URL size	12.8 TB (2016+, striped block blobs) / 1 TB (2012 SP1 CU2–2014, single page blob)
Fabric SQL database (target)	GA. Only the Fabric Migration Assistant is Preview.
Fabric Migration Assistant	Preview tool; DACPAC ≤ 20 MB, on-prem gateway only, no Private Link. These are assistant limits; Fabric SQL database also supports T-SQL, transactional replication, Fabric pipelines / Data Factory copy jobs, Dataflow Gen2, and TDS-capable tools.
Always On AG → SQL VM	SQL Server 2012+ (availability groups start with SQL Server 2012)
Distributed AG → SQL VM	SQL Server 2016+ (distributed availability groups start with SQL Server 2016)
SQL Server 2025 (source)	GA 18 Nov 2025 — native vector/JSON, improved DAG, Entra managed identity when Arc-connected (see below)

NOTE Microsoft Entra managed identity — SQL Server 2025 + Azure Arc only. Connect a SQL Server 2025 instance to Azure Arc and a system-assigned managed identity is created for the SQL Server hostname; associating it with the instance lets SQL Server *"authenticate to Azure services without needing to manage credentials"*. Two directions, and the second is the one that matters for a migration runbook:

- **Inbound** — Entra logins and users connecting *to* SQL Server. Also reachable through an app registration since SQL Server 2022, so this is not the differentiator.
- **Outbound** — SQL Server connecting *to* Azure resources, *"like backup to URL, or connecting to Azure Key Vault"*. An app registration **cannot** do this: outbound needs a primary managed identity. This is the credential-free alternative to the storage-account key or SAS credential that §5 Backup to URL otherwise requires.

Limits, all disqualifying if missed. Arc-enabled **SQL Server 2025 on Windows Server** only — not Linux, not 2022 or earlier. Requires access to the **Azure public cloud**, so it does not apply to the air-gapped and sovereign profiles in §12. **Not supported with failover cluster instances**, which rules out a large share of on-prem estates. Only **system-assigned** identities. Once Entra authentication is enabled, disabling it is not advisable — deleting the registry entries by force *"can result in unpredictable behavior"*.

This is a modernization and credential-hygiene argument for an Arc-connected 2025 estate. It is **not** a migration method, and it does not apply to the older disconnected estates that make up most migration pipelines.

---

10 Cross-cloud sources & reverse migration

Source × method reality check: DMS and LRS cover more cross-cloud sources than MI Link/native restore. MI Link requires SQL Server 2016+, sysadmin on the source, distributed availability group support, ability to create AG endpoints, MI Link network ports (5022 plus MI-side HADR 11000–11999), and network connectivity to the MI VNet; it is therefore not possible from managed-PaaS sources that do not grant sysadmin or custom AG endpoints.

Source	MI Link	LRS	DMS	Native backup/restore	Txn replication	BACPAC/bcp/ADF
On-prem SQL Server / Azure VM	✓ 2016+	✓ 2008–2022	✓	✓ direct `BACKUP TO URL` (2012 SP1 CU2+)	✓ 2016+ publisher	✓
AWS EC2 (SQL on IaaS)	✓ if sysadmin + AG + 5022 + networking	✓ via Blob upload	✓	✓ via Blob upload	✓ if sysadmin	✓
AWS RDS for SQL Server	✗ no sysadmin / no AG endpoints	✓ via S3→Blob upload	✓ offline to Azure SQL DB / MI / VM; ✓ online only to MI / VM (not Azure SQL DB)	indirect only (S3→Blob→restore; no direct `BACKUP TO URL` to Azure)	✗ not practical (requires sysadmin/distributor rights the platform doesn't grant)	✓
GCP Compute Engine (SQL on IaaS)	✓ if sysadmin + AG + 5022 + networking	✓ via Blob upload	✓	✓ via Blob upload	✓ if sysadmin	✓
GCP Cloud SQL for SQL Server	✗ no sysadmin / no AG endpoints	✓ via export→Blob upload	✓	indirect only	✗ not practical (requires sysadmin/distributor rights the platform doesn't grant)	✓

The transactional-replication ✗ for AWS RDS / GCP Cloud SQL is an inference from those platforms' privilege model (no sysadmin / distributor rights), not an explicit Microsoft unsupported-from-RDS statement.

- **Reverse / exit migration:** SQL DB → SQL Server via BACPAC + scripts (heavy); MI → SQL Server 2022 / 2025 via MI Link reverse failback (trivial — a strategic portability argument).
- **Neutral non-Microsoft targets** (for honest comparison): AWS RDS for SQL Server (+ AWS DMS), GCP Cloud SQL (+ GCP DMS), Tessell, OCI self-managed. Azure differentiators: MI Link, Arc, Fabric Mirroring.

---

11 Third-party alternatives (when they beat the native stack)

Tool	Better when…
Striim	Microsoft-recommended **online / CDC** data-migration vehicle for SQL Server → Azure SQL Database (the online path DMS lacks for SQL DB) and for heterogeneous sources (Sybase / Oracle / DB2 → Azure SQL, paired with SSMA for schema); also real-time CDC to Event Hubs / Synapse / Cosmos in parallel.
Dr Migrate / CAST Highlight / UnifyCloud	Portfolio & application/code assessment at scale (wave planning, dependency mapping) — complementary to Azure Migrate.
Qlik Replicate	Heterogeneous sources (Oracle, DB2, iSeries, SAP HANA) with in-flight transforms — more mature than DMS for Oracle→SQL.
Fivetran HVR	Broad CDC + observability; multi-target (Snowflake + Fabric).
Quest SharePlex	SQL Server ↔ Oracle replication; de-Oracle-ization projects.
Carbonite Migrate	VM-level always-on with failback when Azure Migrate/ASR can't handle the OS/hypervisor (KVM, Xen, Citrix).
Veeam / Commvault / Cohesity / Rubrik	Already in place — push backups to Blob, restore on SQL VM (no double licensing).
Redgate / Liquibase / Flyway	Schema versioning (GitOps) post-migration — SqlPackage's weak spot.
VMware HCX (Broadcom)	The zero-downtime "as-is" option for complex VMware clusters incl. SQL FCI → AVS.

---

12 Decision criteria & "when to recommend what"

Criterion	VM	AVS	MI	SQL DB	Fabric SQL DB	AKS / Arc
OS access required	✓	✓	✗	✗	✗	partial
Cross-DB / DTC transactions	✓	✓	–[1]	✗	✗	✓
FILESTREAM / FileTable	✓	✓	✗	✗	✗	✗
PolyBase	✓	✓	–[2]	✗	✗	✓
Service Broker	✓	✓	✓[3]	✗	✗	✓
SQL CLR / linked servers	✓	✓	✓	✗	✗	✓
SQL Agent	✓	✓	✓	✗ (Elastic Jobs)	✗	✓
Min. downtime achievable	near-zero (AG / DAG, planned failover)	~h (vMotion)	~min (MI Link)	min–h	~min with transactional replication (SQL Server 2022 RTM CU12+ publisher, primary keys on replicated tables); otherwise h	depends
Managed patch / upgrade	Auto-patch	✗	✓ Evergreen	✓ Evergreen	✓ Evergreen	✗
Azure Hybrid Benefit	✓	✓	✓	✓ GP/BC vCore provisioned; ✗ DTU/serverless/Fabric SQL DB; Hyperscale single DBs with provisioned compute **created before 15 Dec 2023** only, and only until Dec 2026	n/a	✓
Sovereignty / edge	✓	✓	limited regions	limited regions	limited regions	✓ (Arc)

[1] Azure SQL MI supports SQL-to-SQL distributed transactions between SQL MIs and SQL Server / SQL Server-based products via managed DTC (enable it and open port 135 **both inbound and outbound**, 14000–15000 inbound, 49152–65535 outbound, in both the MI subnet NSG and any external firewall). DTC to third-party RDBMS and Azure SQL Database is not supported; if all participants are SQL MI, prefer native elastic transactions. [2] Azure SQL

MI supports data virtualization over Azure Blob Storage / ADLS Gen2 for CSV/delimited and Parquet via `OPENROWSET(BULK)`, `CREATE EXTERNAL TABLE` and CETAS. It does not support Delta Lake, S3-compatible storage, push-down computation, or PolyBase connectors to external RDBMS (Oracle, Teradata, MongoDB, another SQL Server). [3] Azure SQL MI supports Service Broker **within a single instance** as GA. Cross-instance message exchange, MI-to-MI and SQL Server-to-MI, is in **public preview**: `CREATE ROUTE` and `ALTER ROUTE` must specify port 4022, transport security only, `CREATE REMOTE SERVICE BINDING` unsupported. Treat it as a preview-gated capability, not a hard MI blocker.

Client profile	Neutral recommendation	Why
Banking / regulated / strong on-prem	SQL VM + ESU via Arc, or AVS	OS control, compliance, soft transition
Multi-tenant SaaS ISV	SQL DB Hyperscale / Elastic Pool	Per-tenant elasticity, cost control
Heavy legacy ERP (SAP, on-prem Dynamics)	SQL MI or SQL VM	Instance & SQL Agent compatibility
Modern micro-services app	SQL DB serverless	Pay-per-use, auto-pause
Fabric-native / analytics-first	SQL DB in Fabric + Mirroring	OLTP + OneLake unification
Edge / sovereign / multi-cloud	Arc-enabled SQL MI on local AKS	Sovereignty + Azure consistency
Short-term data-center exit	AVS + VMware HCX	Zero refactor, vMotion
Modernize off end-of-support	SQL VM "as-is" + ESU boundary check	The ESU programme covers SQL Server 2014 and 2016 only; free ESU on Azure VM/AVS applies to SQL Server 2014, while SQL Server 2016 needs paid ESU even on Azure VM

---

13 Field insights — recurring pitfalls

- **TDE certificate**: protect-then-restore order matters — a TDE-protected database restore fails on the destination unless the TDE certificate / asymmetric key is first installed in the destination server's `master` database.
- **Windows logins**: DMS skips them by default; enable the option and grant MI read to Entra ID (Privileged Role Administrator).
- **MI Link ports**: MI subnet NSG needs inbound 5022 and 11000–11999 from SQL Server plus outbound 5022; SQL Server host/corporate firewalls need inbound 5022 from the MI subnet and outbound 5022 and 11000–11999 to MI. The 11000–11999 range is always required for MI-side HADR data replication and is frequently missed in locked-down networks.
- **Other methods need network too**: DMS / LRS / transactional replication require outbound HTTPS (443) to Azure Storage/Blob, SQL 1433 (and 1434/UDP SQL Browser for named instances) — anticipate firewall/NSG blocks in locked-down environments.
- **DAG**: requires AD Domain Services (or workgroup AG + certs) — an infra blocker architects forget.
- **Retained server name via DNS redirect to MI**: a common pattern is to keep the old server name and repoint DNS at Managed Instance. Clients that validate the TLS hostname, or that set `HostNameInCertificate`, can break at cutover because the certificate presented by MI is not the one they expect — and Microsoft

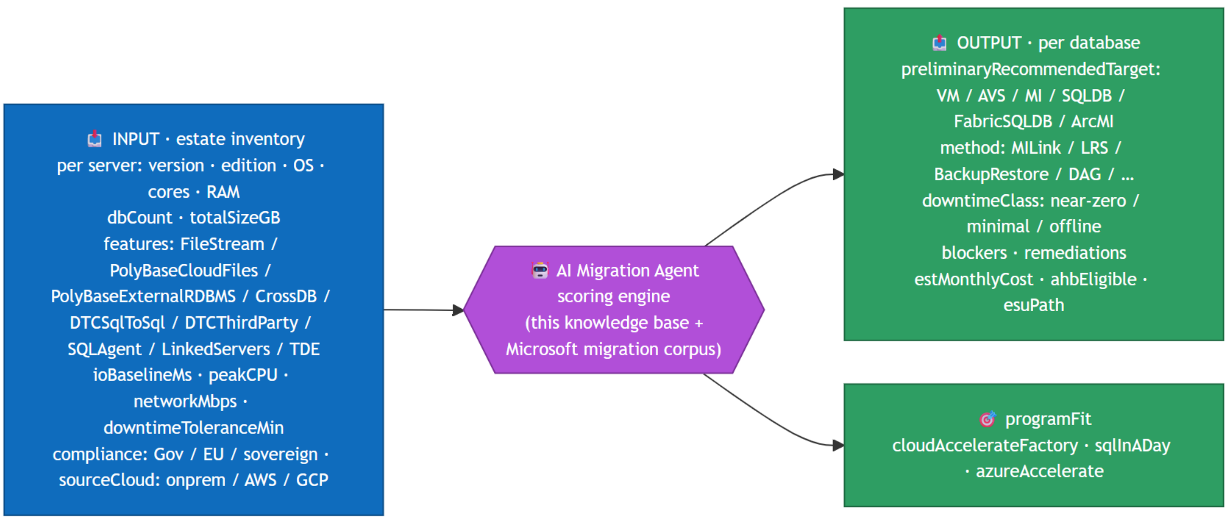
is changing the instance certificate used for exactly this pattern. Inventory those clients, test them against the target MI certificate **before** the DNS change, and update client settings for the new behaviour. Do not assume the retained name will validate. HostNameInCertificate changes in Azure SQL MI

- **Transactional replication → SQL DB / Fabric SQL DB / MI**: Azure SQL Database subscribers require SQL Server 2016+ publishers; Fabric SQL database subscribers require SQL Server 2022 RTM CU12+ publishers and do not support Private Link for replication; Azure SQL MI subscribers require SQL Server 2016+ publishers, with exact combinations dependent on MI update policy. Snapshot and one-way transactional replication are supported for SQL DB/Fabric; peer-to-peer and merge are not; replicated tables require primary keys; distribution DB and agents cannot live in Azure SQL Database.
- **Hyperscale**: the only viable SQL DB choice above 4 TB or with heavy concurrent write I/O — and it stops at **128 TB** for a single database (inside a Hyperscale elastic pool the per-database maximum is **100 TB**). A single database beyond that cannot be rehosted as one database: Azure SQL MI tops out far below that ceiling, so it is **not** an as-is destination. Shard across databases or instances, or use SQL Server on Azure VM subject to its own storage design and limits.
- **Fabric SQL DB**: the *target* is GA; only the Migration Assistant is Preview, with a 20 MB DACPAC cap, on-prem gateway only and no Private Link. Those tool limits are not target-wide limits, because T-SQL, transactional replication, Fabric pipelines / Data Factory copy jobs, Dataflow Gen2 and TDS-capable tools can also ingest data — so don't eliminate the target because the assistant does not fit. Do assess the database surface: it is a subset of the SQL Server surface.
- **Mirroring ≠ migration**: it's continuous analytics replication; treating it as a one-shot migration is a dangerous shortcut.
- **Dependency mapping**: undocumented linked servers and SQL Agent jobs are among the most commonly late-discovered blockers — run a dependency map (Azure Migrate or third-party) before committing a target.
- **SLAs differ**: MI BC 99.99% · SQL DB BC 99.995% (zone-redundant) · SQL DB Hyperscale 99.99% · SQL VM depends on the AG. Align to the app, not a slogan.
- **Security by design**: SQL DB has a public endpoint by default — recommend Private Endpoint + Entra-only auth + disabled SQL auth from day one.
- **Land the zone first**: stand up an Azure **landing zone** (CAF — IAM, policy, networking, monitoring, Defender pre-baked) before migrating. Once deployed, each workload move stops being a one-off because identity, policy, networking, monitoring and Defender are already in place.
- **Fabric SQL DB ≠ Azure SQL DB**: not a drop-in replacement — fine-grained security (RLS / OLS / dynamic masking) does not propagate to OneLake; governance must be re-implemented in Fabric. For analytics on operational data, prefer Mirroring (keep OLTP on Azure SQL, mirror into Fabric, no ETL) over a hard cutover.
- **Bootstrap via Arc first**: connect the estate to Azure Arc *before* migrating — free weekly continuous assessment, blocker detection, ESU and PAYG (OpEx) licensing while you plan, with zero workload change (§3 row 8).

14 FY27 SQL Motion context & AI Migration Agent

This document is the knowledge base behind the **FY27 EMEA EPS — Data Motion "SQL in a Day"**. Two notes:

- **Storyline correction.** The deck line *"Azure Migrate, DMA, DMS, SSMA, Cloud Accelerate Factory"* must drop DMA (retired) → use *"Azure Migrate, SSMS 22 + Arc-based assessment, DMS, SSMA, Cloud Accelerate Factory"*. A 4th target pillar — SQL Server enabled by Azure Arc — should appear alongside VM / MI / SQL DB.
- **AI Migration Agent — I/O contract.** The afternoon agent scores a preliminary recommended assessment path for the customer estate; confirm outputs with tool-based assessment and an architect. Suggested flow so it's usable by both humans and automation:



Microsoft programs to attach: Cloud Accelerate Factory (zero-cost delivery), Azure Accelerate / FastTrack, AHB + ESU via Arc — detailed in §15.

---

15 Commercial levers & funding programs (FY27)

Two money levers to combine on every deal: **(A) Microsoft-funded engagement** money (assess / pilot / migrate) and (B) commercial / licensing levers that durably cut TCO. **Amounts, ratios and program names are partner-confidential and drift each fiscal year (FY27 starts 1 Jul 2026) — re-validate live in Partner Center / the MCI guide before quoting.**

15.1 Commercial / licensing levers (permanent TCO reducers)

Lever	What it saves	Mechanics
Azure Hybrid Benefit (AHB)	30%+ on eligible Azure SQL DB / MI; compute-only on VM	License + Software Assurance reallocated to Azure; 1 Enterprise core = 4 GP vCores; 180-day dual-use during migration; portal toggle. Applies to Azure SQL Database General Purpose / Business Critical in the vCore provisioned compute tier, not to the DTU model, serverless compute tier, or Fabric SQL DB. Hyperscale carries a creation-date cohort: AHB can only be applied to Hyperscale single databases with provisioned compute **created before 15 December 2023**, and only until December 2026, after which they too move to the simplified pricing. Hyperscale databases created on or after that date are **not** AHB-eligible because the simplified pricing already removed the software licence fee.
ESU on Azure — version boundary	Removes ESU cost only for eligible older versions	The SQL Server ESU programme now covers **SQL Server 2014 and SQL Server 2016 only**: 2014 reached end of support on 10 July 2024 with ESUs until 13 July 2027, and 2016 reached end of support on 15 July 2026 with ESUs until 17 July 2029. Azure VMs / AVS: ESU free and automatic for SQL Server 2014; SQL Server 2016 requires a paid ESU subscription even on Azure VM. SQL Server 2012 and earlier have no ESU path left, so do not describe them as covered. Azure SQL PaaS has no ESU concept; on-prem / multicloud / hosted use Azure Arc for ESU subscription or PAYG billing; non-prod free when prod runs ESU via Arc.
PAYG licensing via Arc	Turns the SQL license into OpEx	Billed only when SQL runs; CALs included; can sit inside a MACC. Requires active SA or PAYG enabled.
Free Azure SQL offers	Zero-cost POC / pilots	MI free 12 months; SQL DB free for the subscription lifetime (serverless GP).
Reservations / Savings Plans	1- / 3-yr commitment discount	Stacks with AHB. Partner Earned Credit (15%) does not apply to reservations.
Savings plan for databases	up to 35% on Azure SQL (DB / MI)	1- or 3-year hourly compute commitment; auto-applies across participating database services up to the commitment; stacks with AHB.

15.2 Microsoft-funded engagement programs

Program	Funds	Access
Azure Accelerate (FY26 umbrella = *Azure Migrate & Modernize* + *Azure Innovate* + *Cloud Accelerate Factory*)	Assessments, POC, pilots, deployments, Azure credits, skilling	Self-serve — Partner Center nomination → POE → paid ≤ 45 days
Azure Accelerate for Databases	SQL / data-estate modernization	Same workflow (database scenario)
Cloud Accelerate Factory	Zero-cost Microsoft expert deployment	Via Azure Accelerate in Partner Center
FastTrack for Azure	Free Microsoft engineering guidance (not cash)	Account team / program alias
ECIF (End-Customer Investment Funds)	Assessments, POCs, migrations, training	Field-led via PDM / AE — not self-serve
Azure Frontier Offer (ECIF + ACO, dual-run offset)	Services + dual-run during migration; competitive Oracle / legacy displacement	Field-led via PDM; typically ≥ 50% ACR from Fabric / first-party Databases / AI

15.3 Partner mechanics

- **Attribution first** (funding prerequisite): link the customer via PAL (Partner Admin Link) or DPOR → unlocks the 15% Partner Earned Credit on managed Azure consumption.
- **Routes:** *Self-serve* (Partner Center) — Azure Accelerate, Databases, Cloud Accelerate Factory · *Field-led* (engage the PDM early) — ECIF, Azure Frontier Offer · *Customer self-serve* (Azure portal) — AHB, ESU, free offers, reservations.
- **Typical sequence:** Azure Migrate discovery + TCO → nominate assess/pilot in Azure Accelerate (Databases) → if competitive or ≥ 50% data ACR, engage PDM for the Azure Frontier Offer → pilot on free SQL DB / MI → CAF for zero-cost deploy → POE → lock run-rate with reservations.

16 Sources (Microsoft Learn)

Overviews & taxonomy - Database Migration hub — https://learn.microsoft.com/en-us/data-migration/ - SQL Server → Azure SQL Database — https://learn.microsoft.com/en-us/data-migration/sql-server/database/overview - SQL Server → Azure SQL Managed Instance — https://learn.microsoft.com/en-us/data-migration/sql-server/managed-instance/overview - SQL Server → SQL Server on Azure VM — https://learn.microsoft.com/en-us/data-migration/sql-server/virtual-machines/overview - Azure SQL feature comparison — https://learn.microsoft.com/en-us/azure/azure-sql/database/features-comparison - PolyBase data virtualization guide — https://learn.microsoft.com/en-us/sql/relational-databases/polybase/data-virtualization-guide - SQL MI data virtualization overview — https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/data-virtualization-overview - SQL MI distributed transaction coordinator (DTC) — https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/distributed-transaction-coordinator-dtc

Control planes / tools - Azure Migrate (SQL assessment) — https://learn.microsoft.com/en-us/azure/migrate/how-to-create-azure-sql-assessment - SQL Server migration in Azure Arc — https://learn.microsoft.com/en-us/sql/sql-server/azure-arc/migration-overview - SQL migration to SQL Server on Azure VMs in Azure Arc (GA how-to) — https://learn.microsoft.com/en-us/sql/sql-server/azure-arc/migrate-to-sql-server-on-azure-vms?view=sql-server-ver17 - Migrate to Azure SQL Managed Instance in Azure Arc — https://learn.microsoft.com/en-us/sql/sql-server/azure-arc/migrate-to-azure-sql-managed-instance - GA announcement — SQL migration to SQL Server on Azure Virtual Machines in Azure Arc — https://techcommunity.microsoft.com/blog/microsoftdatamigration/generally-available-sql-migration-to-sql-server-on-azure-virtual-machines-in-azu/4536940 - Managed identity overview — SQL Server enabled by Azure Arc — https://learn.microsoft.com/en-us/sql/sql-server/azure-arc/managed-identity?view=sql-server-ver17 - Set up managed identity and Microsoft Entra authentication — SQL Server enabled by Azure Arc — https://learn.microsoft.com/en-us/sql/sql-server/azure-arc/microsoft-entra-authentication-with-managed-identity?view=sql-server-ver17 - SSMS (download/overview) — https://learn.microsoft.com/en-us/sql/ssms/sql-server-management-studio-ssms - Azure Database Migration Service — https://learn.microsoft.com/en-us/azure/dms/dms-overview - DMS supported scenarios (offline/online per target) — https://learn.microsoft.com/en-us/azure/dms/resource-scenario-status - `Az.DataMigration` PowerShell — https://learn.microsoft.com/en-us/powershell/module/az.datamigration/ - SSMA — https://learn.microsoft.com/en-us/sql/ssma/sql-server-migration-assistant - Database Experimentation Assistant (DEA) — https://learn.microsoft.com/en-us/previous-versions/sql/dea/database-experimentation-assistant-overview - SQL Server Distributed Replay — https://learn.microsoft.com/en-us/sql/tools/distributed-replay/sql-server-distributed-replay - RML Utilities / OStress — https://learn.microsoft.com/en-us/troubleshoot/sql/tools/replay-markup-language-utility - What's happening with Azure Data Studio — https://learn.microsoft.com/en-us/sql/tools/whats-happening-azure-data-studio

Methods - Managed Instance link — https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/managed-instance-link-feature-overview - Managed Instance link preparation — https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/managed-instance-link-preparation - Log Replay Service — https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/log-replay-service-migrate - Log Replay Service overview — https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/log-replay-service-overview - Compare LRS and MI Link — https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/log-replay-service-compare-mi-link - Native backup & restore (MI) — https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/restore-sample-database-quickstart - Transactional replication (SQL DB) — https://learn.microsoft.com/en-us/azure/azure-sql/database/replication-to-sql-database - Import/Export · BACPAC — https://learn.microsoft.com/en-us/azure/azure-sql/database/database-import - bcp utility — https://learn.microsoft.com/en-us/sql/tools/bcp/bcp-utility - Connect to SQL database in Fabric with bcp — https://learn.microsoft.com/en-us/fabric/database/sql/connect#connect-with-bcp-utility - Fabric Data Factory SQL database connector — https://learn.microsoft.com/en-us/fabric/data-factory/connector-sql-database-overview - Smart Bulk Copy — https://github.com/Azure-Samples/smartbulkcopy - Backup to URL — https://learn.microsoft.com/en-us/sql/relational-databases/backup-restore/sql-server-backup-to-url - Move a TDE-protected database — https://learn.microsoft.com/en-us/sql/relational-databases/security/encryption/move-a-tde-protected-database-to-another-sql-server - Availability group → Azure VM — https://learn.microsoft.com/en-us/data-migration/sql-server/virtual-machines/availability-group-migrate - Log shipping — https://learn.microsoft.com/en-us/sql/database-engine/log-shipping/about-log-shipping-sql-server - Elastic Jobs (SQL DB) — https://learn.microsoft.com/en-us/azure/azure-sql/database/elastic-jobs-overview

Targets — containers, Fabric, AVS, Arc - Create Arc-enabled SQL Managed Instance — https://learn.microsoft.com/en-us/azure/azure-arc/data/create-sql-managed-instance - Migrate to Arc-enabled SQL MI — https://learn.microsoft.com/en-us/azure/azure-arc/data/migrate-to-managed-instance - SQL Server on Kubernetes / AKS — https://learn.microsoft.com/en-us/sql/linux/quickstart-sql-server-containers-kubernetes - SQL Server in a Docker container — https://learn.microsoft.com/en-us/sql/linux/quickstart-install-connect-docker - Fabric Migration Assistant for SQL database (Preview) — https://learn.microsoft.com/en-us/fabric/database/sql/migration-assistant - Ingest data into Fabric SQL database — https://learn.microsoft.com/en-us/fabric/database/sql/tutorial-ingest-data - Migrate to Fabric SQL DB via DACPAC — https://learn.microsoft.com/en-us/fabric/database/sql/migrate-with-migration-assistant-using-dacpac - Fabric Mirroring for SQL Server — https://learn.microsoft.com/en-us/fabric/mirroring/sql-server - Limitations for SQL database in Fabric — https://learn.microsoft.com/en-us/fabric/database/sql/limitations - Fabric — choose a data store (decision guide) — https://learn.microsoft.com/en-us/fabric/fundamentals/decision-guide-data-store - Azure VMware Solution — https://learn.microsoft.com/en-us/azure/azure-vmware/introduction - SQL MI — Next-gen General Purpose — https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/service-tiers-next-gen-general-purpose-use - Azure SQL DB Hyperscale — https://learn.microsoft.com/en-us/azure/azure-sql/database/service-tier-hyperscale - Reporting Services consolidation FAQ — https://learn.microsoft.com/en-us/sql/reporting-services/reporting-services-consolidation-faq - Migrate SSRS reports to Power BI — https://learn.microsoft.com/en-us/power-bi/guidance/migrate-ssrs-reports-to-power-bi

Data movement & VM-level - Azure Data Box / Data Box Heavy — https://learn.microsoft.com/en-us/azure/databox/data-box-overview - ExpressRoute — https://learn.microsoft.com/en-us/azure/expressroute/expressroute-introduction - AzCopy — https://learn.microsoft.com/en-us/azure/storage/common/storage-use-azcopy-v10 - Azure Storage Mover — https://learn.microsoft.com/en-us/azure/storage-mover/service-overview - Azure Site Recovery — https://learn.microsoft.com/en-us/azure/site-recovery/site-recovery-overview - Azure-SSIS Integration Runtime — https://learn.microsoft.com/en-us/azure/data-factory/create-azure-ssis-integration-runtime - SQL Data Sync (retiring 30 Sep 2027) — https://learn.microsoft.com/en-us/azure/azure-sql/database/sql-data-sync-data-sql-server-sql-database?view=azuresql

Licensing / programs / lifecycle - Azure Hybrid Benefit (Azure SQL) — https://learn.microsoft.com/en-us/azure/azure-sql/azure-hybrid-benefit - SQL Server ESU enabled by Azure Arc — https://learn.microsoft.com/en-us/sql/sql-server/azure-arc/extended-security-updates - Move SQL Server license to PAYG (Arc) — https://learn.microsoft.com/en-us/sql/sql-server/azure-arc/manage-pay-as-you-go-transition - SQL Server end-of-support / ESU — https://learn.microsoft.com/en-us/sql/sql-server/end-of-support/sql-server-extended-security-updates - SQL Server 2016 lifecycle — https://learn.microsoft.com/en-us/lifecycle/products/sql-server-2016 - Azure SQL MI free offer — https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/free-offer - Azure SQL DB free offer — https://learn.microsoft.com/en-us/azure/azure-sql/database/free-offer - Azure Accelerate (partner programs) — https://partner.microsoft.com/en-us/partnership/azure-offerings - Tool consolidation / retirement (blog) — https://www.microsoft.com/en-us/sql-server/blog/2024/09/12/modernize-your-database-with-the-consolidation-and-retirement-of-azure-database-migration-tools/ - Cloud Adoption Framework — migrate — https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/migrate/ - Azure landing zones (CAF) — https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/landing-zone/ - Savings plan for databases — https://learn.microsoft.com/en-us/azure/cost-management-billing/savings-plan/savings-plan-overview - Modernize your databases (hub) — https://aka.ms/modernizedatabases

> *Links last verified: 5 August 2026. Microsoft migration guides moved from `…/azure-sql/migration-guides/…` to `…/data-migration/sql-server/…` (redirects in place). Items marked **Preview** are subject to change.*

---

17 Document version & changelog

Current version: **v2.9** (2026-08-17).

Version history (current: v2.9)

Version	Date	Changes
v2.9	2026-08-17	**Three documents disagreed about what happens when a prerequisite is unknown, and the disagreement decided what the tool recommends.** `MI-LINK-HOST` said in the rule index that an unknown host or edition **refuses** MI Link, while §B3 and the executable mirror both treated it as `unknown_requires_assessment` — refusing on absence of evidence makes an information gap look like an incompatibility. `BACKUP-BLOB-PATH` claimed every native backup/restore variant moves through Azure Blob, which is true for Managed Instance and false for a VM: a profile that required a VM lost its recommendation to a shortlist when Blob was blocked, although this document documents backup to a file and a copy into the target. And FILESTREAM grouped the Linux container with the VM as eligible, offering a target that has no FILESTREAM at all. A new `rule-unknown-behaviour-agrees` gate now compares the rule index, the section that defines each rule and the input contract, so a contradiction of this kind fails the build instead of waiting for a model to notice it.
v2.8	2026-08-14	**The rule index pointed 26 of its 28 rules at sections that never mentioned them, and three at the wrong section outright.** Every recommendation cites a rule ID, and the index is how a reader turns that ID into the text they can argue with. `FABRIC-TARGET` and `FABRIC-ASSISTANT` both addressed A2, the hard compatibility table, which contains no Fabric text; the Fabric branch is step 4 of A3 and the assistant limits live in B3. `HYPERSCALE-CEILING` addressed `A2, B2` when the 128 TB ceiling is stated only in B2, and `SOURCE-PERMISSIONS` addressed B3 alone while the input it gates is normalized in A0. The remaining 22 pointers named a real section that carried no trace of the rule, so following one led to a page of prose with nothing to match against. Rule IDs are now anchored in the text they govern across A0, A2, A3, A4, B1, B2, B3, C1 and C4, and the five wrong or incomplete addresses are corrected. No recommendation changes: the skill loads the whole policy document, so the normative text always reached the model regardless of what the index said, and the effect column beside each address was already correct. What was broken was traceability. The `rule-index-consistent` gate caused this by reading four groups from a five-column table, which left the `Defined in` column checked by nothing for five releases; it now resolves every pointer to a section that exists and mentions the rule.
v2.7	2026-08-13	**The summary matrix asserted two things Microsoft's own documentation contradicts, and both were wrong in the direction that costs a customer a working route.** §8 gave `bcp / Smart Bulk Copy` a – against **SQL database in Fabric**. bcp names *SQL database in Microsoft Fabric* in its own **Applies to** banner, and Fabric publishes a dedicated Connect with bcp utility procedure — "just like any other SQL Database Engine product". The cell is now `✓`, with the constraint that makes it work recorded: Fabric SQL database accepts no SQL authentication, so the connection must use Microsoft Entra ID with `-G`. The same row also fused **two tools with different support**, and that fusion was what hid the error — it claimed Arc SQL MI and containers for Smart Bulk Copy, an archived community sample that claims neither. bcp and Smart Bulk Copy are now separate rows. Second, `ADF Copy` claimed **Fabric SQL DB**. Azure Data Factory publishes Fabric **Lakehouse** and Fabric **Warehouse** connectors and **no Fabric SQL database connector**; the product that has one is **Fabric** Data Factory, whose SQL database connector is Beta and serves Copy activity, Copy job and Dataflow Gen2 in both directions. The row is renamed `Data Factory Copy` and the distinction is stated in the legend and in §5.4, because a reader who built an Azure Data Factory pipeline against this target would find no connector to select. 56 supported cells, up from 51.
v2.6	2026-08-12	**One addition, from an external review of the knowledge base against public Microsoft sources.** Microsoft Entra **managed identity** was absent entirely, and it is the only capability in that review the document did not already carry. It is recorded where a reader looks up what a source version buys them (§9), scoped exactly as the sources scope it — Arc-connected **SQL Server 2025 on Windows Server**, system-assigned only. The migration-relevant half is the **outbound** direction: an app registration cannot make outbound connections, so a managed identity is what lets `BACKUP TO URL` and Key Vault access work without a storage-account key or SAS credential. The limits are recorded with the same weight as the capability, because each one disqualifies it outright: no Linux, no SQL Server 2022 or earlier, no failover cluster instances, and Azure public cloud required — which excludes the air-gapped and sovereign profiles in §12.
v2.5	2026-08-12	**A knowledge-base fact changed, and it was a rule that contradicted this document's own table.** The AzCopy row stated a blanket "*for DBs > 1 TB, local backup + AzCopy is faster/safer than direct Backup-to-URL*", while the Backup to URL row four sections earlier records the real constraint: 1 TB is the **page-blob cap on SQL Server 2012 SP1 CU2–2014**, and SQL Server 2016+ writes block blobs reaching **12.8 TB striped**. Read alone, the AzCopy row pushed a reader on a supported 2019 source off a direct path for a limit that does not apply to it. The row now scopes the cutoff to the builds it belongs to and states that above them the choice is throughput and retry behaviour, not a size limit. Found by the weekly review.
v2.4	2026-08-12	**A factual correction, and the first knowledge-base fact to change since v1.18.** SQL Server 2014 was documented with ESUs "*until 8 July 2027*". Microsoft Lifecycle records Extended Security Updates Year 3 for SQL Server 2014 ending **13 July 2027** — a five-day error on a date that drives stay-versus-migrate economics for the estates most likely to be assessed. The end-of-support dates for 2014 and 2016 also used the Patch Tuesday convention (9 July 2024, 14 July 2026) while Microsoft publishes the Extended End Date (10 July 2024, 15 July 2026). Both are defensible in isolation and contradictory side by side, which is exactly how a customer conversation goes wrong. The Lifecycle dates are now quoted as published, and a note records why the Patch Tuesday differs, so the two never look like a mistake again. Found while verifying a third-party audit that had raised the 2016 date as a nuance; the 2014 error it did not see was the one that mattered. Applying the convention then exposed the same off-by-one in three further rows, all quoting the Patch Tuesday: SQL Server 2017 (12 -> 13 Oct 2027), 2019 (8 -> 9 Jan 2030), 2022 (11 -> 12 Jan 2033), and SSRS 2022, which inherits the SQL Server 2022 lifecycle. Every row of the table is now verified against its Microsoft Lifecycle page, and the table links to those pages so a reader can check the dates without trusting us.
v2.3	2026-08-11	**No knowledge-base fact changed. Four rules that were written and applied nowhere are now executed.** An interactive graph of the rule set made them visible: each appeared in the index, was described in the prose, and decided nothing. **`HYPERSCALE-CEILING`** was the one that could mislead a customer — a database above the 128 TB single-database ceiling was recommended onto Azure SQL Database at medium confidence with its method gate reported as passed. Past that ceiling neither PaaS family holds the workload as it stands, so both are refused and the answer becomes SQL Server on Azure VM with sharding named as the alternative. **A refused gate now removes its method:** printing `refused` beside the method it refused left the card contradicting itself. **`SOURCE-PERMISSIONS`** is consumed — MI Link configures an availability group and a publisher needs its own rights, so limited rights refuse the method and unstated rights hold it at unknown. **`LRS-WINDOW`** is executed: the 30-day maximum existed only as a guard scenario in the data file, and the constraint is now always recorded, becoming an unknown once size or bandwidth make 30 days a real risk. Ordering caught us a third time and a gate caught it: these gates first ran before the consistency pass, judging a method that pass then replaced. They run after it now, and findings raised against a rejected method are dropped with it. The rule count had drifted to 26 in five documents since v2.1, so CI checks it alongside gates, scenarios and invariants. 23 gates, 110 scenarios, 28 addressable rules.
v2.2	2026-08-11	**No knowledge-base fact changed. The skill is renamed `recommend-migration-path`, and the stamp moves so the skill, the rules and this document stay pinned to one commit.** The name `get-migration-assessment` is already taken, by a skill in `microsoft/sql-migration-agent` that does something else entirely: it reads assessment results from Azure Resource Manager for Arc-enabled instances, requiring `az login` and an assessment that already exists. This skill interviews a person precisely when no such data exists. Two skills doing different jobs under one name is not a cosmetic problem — anyone installing both plugins would have had two entries called `get-migration-assessment` in `/skills`, with no way to tell which answered. The new name says what this one produces. It also makes the contribution a straight copy rather than a rename applied by hand on every refresh, which is the kind of manual step that gets forgotten. The two skills are complementary and now say so: this one routes to `get-migration-assessment` when an assessment already exists, because measured evidence beats an interview every time.
v2.1	2026-08-10	**No knowledge-base fact changed. The stamp moves so the skill, the rules and this document stay pinned to one commit.** A second external audit scored v2.0 at 8/10 and found the gap v2.0 had not closed: the contracts were written, but nothing proved the **interview** obeys them. A real session answered in option IDs the contract had never heard of — six of ten sampled — and collected eight fields it never declared, while every gate stayed green. **Contract.** 72 option IDs instead of 30: every enum now has stable IDs, and the missing fields are declared, including `rpo`, `rto`, `clr_permission_set`, `tde_status`, `source_permissions`, `authentication` and `blob_https_reachability`. A new gate checks the direction nobody was checking, from the interview to the contract. **Network.** One question became three. Bandwidth, MI Link ports (5022 and 11000–11999) and Blob reachability gate different things, and merging them let a session report a backup/restore gate as `passed` while the upload path was never verified. **Rules that existed only on paper are now implemented.** `BACKUP-BLOB-PATH` holds any backup-based method — including Log Replay Service, which stages its backups in Blob — at `unknown_requires_assessment` until the path is confirmed. `CLR-PERMISSION` is written normatively and applied: `UNSAFE` or an unstated permission set returns a shortlist rather than a confident Managed Instance recommendation, and `SAFE` is not a clearance because `clr strict security` treats SAFE and EXTERNAL_ACCESS as UNSAFE without a signature. **Size classes no longer overlap**, so a 200 TB database no longer matches two answers. **Four invariants added**, 9 to 13: a method gate may not pass on an unknown, all eight target families must appear in the trace, `excluded_by_preference` is distinct from `unsupported`, and `normalizedProfile` must be present. **The live knowledge-base URL was returning 404**: it carried `/blob/`, so the fetch the README advertised had never worked. **The skill now announces what it loaded** and checks once whether a newer release exists. 106 scenarios, 23 gates.
v2.0	2026-08-10	**No knowledge-base fact changed. The version stamp moves so the skill, the rules and this document stay pinned to one commit.** A second external audit went after the design rather than the facts: the repository tested what was easy to test rather than what makes the decision. Waves 0 to 2 answered the factual findings in v1.18; v2.0 answers the structural ones. **Contracts.** The interview's vocabulary and the answer's shape were implicit, so they drifted — the defect that made a user who answered *no dependencies* read *dependencies unknown* was a missing distinction, not a broken rule. `reference/input-contract.md` now owns the 30 option IDs, the 20 canonical fields and the three states an answer can hold: confirmed-none, unknown, and not-applicable. `reference/output-contract.md` owns the status vocabulary and 9 invariants. **The skill checks its own answer.** Before the card is shown it re-reads its draft against those invariants

Versioning: **MAJOR.MINOR** — MINOR bumps for fact/link refresh and additive content; MAJOR bumps for structural rewrites or scope changes.

18 Appendix — Using the `sql-migration-advisor` Copilot CLI skill

This knowledge base also powers `sql-migration-advisor`, a GitHub Copilot CLI skill that turns the inventory above into a guided, preliminary recommendation. Ask Copilot "*migrate a SQL Server environment to Azure*": it runs a short two-tier interview — triage (scope, source location & version, intent, feature dependencies, size, downtime tolerance, network, compliance, ancillary services), then confirmation questions only where they can change the answer — and returns a grounded recommendation card: target and tier, method, downtime class, blockers + remediations, the best alternative, confidence and open evidence gaps, cost levers, and the right Microsoft program. It never recommends retired tooling (DMA, Azure Data Studio extension, DMS classic).

18.1 Install (once)

macOS / Linux:

```
git clone https://github.com/fredgis/sql-migration-advisor.git
↪ ~/.copilot/skills/sql-migration-advisor
```

Windows (PowerShell):

```
git clone https://github.com/fredgis/sql-migration-advisor.git
↪ "$env:USERPROFILE\.copilot\skills\sql-migration-advisor"
```

Restart Copilot CLI, run `/skills` to confirm `sql-migration-advisor` is listed, then ask your question in natural language (English or French).

18.2 Example output

The card below is a real recommendation for an estate of ~100 on-premises SQL Server 2012 databases (limited WAN, exiting out-of-support):